ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** reflection	Reflection.
2	**(d)** convex	Bulges.
3	**(a)** focal length	f.
4	**(b)** angle of reflection	Law.
5	**(a)** concave	Hollow.
6	**(d)** 10 cm	R/2.
7	**(d)** perpendicular to the surface	Right angle.
8	**(c)** converge	Converge.
9	**(d)** 16 cm	2f.
10	**(c)** virtual	Behind the mirror.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bouncing back of light from a surface (2).	2
12	Image that cannot be obtained on a screen (2).	2
13	Concave mirror (2).	2
14	Angle of incidence equals angle of reflection (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Virtual (1); Erect (1); Same size (1).	1+1+1
16	n = 360/120 - 1 (1); = 3 - 1 (1); = 2 (1).	1+1+1
17	It diverges rays (1); Image falls on retina (1); Corrects myopia (1).	1+1+1
	OR: Dressing (1); Periscope (1); Kaleidoscope (1).	1+1+1
18	f = R/2 (1); 40/2 (1); 20 cm (1).	1+1+1
	OR: Two parallel mirrors (1); Each at 45 degrees (1); Light reflects twice (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Statement of law 1 (1); Statement of law 2 (1); Pins and paper (1); Measure angles (1); i = r (1).	1+1+1+1+1
	OR: Regular: mirror (1); Diffuse: paper (1); Both obey laws (1); Image only in regular (1); We see objects due to diffuse reflection (1).	1+1+1+1+1
20	Virtual (1); Erect (1); Same size (1); Distance equal (1); Laterally inverted (1).	1+1+1+1+1
	OR: Candle, lens, screen (1); Adjust distance (1); Inverted image (1); Real image (1); Size changes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** diverge	Spread.
2	**(a)** 5 cm behind the mirror	Same distance.
3	**(c)** equal to object	Same.
4	**(c)** convex lens	Magnifies.
5	**(a)** diffuse reflection	Scattering.
6	**(c)** plane	Flat.
7	**(c)** concave lens	Myopia.
8	**(a)** regular reflection	Mirror.
9	**(c)** concave	Converging.
10	**(c)** convex lens	Real image.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Concave (1); Convex (1).	1+1
12	Magnifying glass (1); Camera (1).	1+1
13	A line perpendicular to the surface at the point of incidence (2).	2
14	Convex (1); Concave (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Shaving (1); Dentist (1); Headlight reflector (1).	1+1+1
16	R = 2f (1); 2 x 12 (1); 24 cm (1).	1+1+1
17	Diffuse reflection (1); Rays scatter (1); No image (1).	1+1+1
	OR: Rear-view (1); Street corners (1); Shops (1).	1+1+1
18	Screen (1); Inverted/erect (1); Formed by actual or apparent rays (1).	1+1+1
	OR: Focus Sun's image on a screen (1); Measure lens-to-screen distance (1); It is f (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Surface (1); Image formed (1); Concave uses (1); Convex uses (1); Field of view (1).	1+1+1+1+1
	OR: Pole (1); C (1); Axis (1); Focus (1); f = R/2 (1).	1+1+1+1+1
20	Three mirrors (1); At 60 degrees (1); Coloured beads (1); Multiple reflections (1); Patterns (1).	1+1+1+1+1
	OR: Rear-view convex (1); Shops use convex (1); Wide field (1); Plane in dressing (1); Headlights concave (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** screen	Real.
2	**(a)** convex	Diverging.
3	**(b)** two plane mirrors	Submarine use.
4	**(c)** left-right reversal	Mirror.
5	**(a)** convex	Wide view.
6	**(a)** 3	n = 4 - 1.
7	**(c)** straight lines	Rectilinear.
8	**(c)** concave	Enlarged image.
9	**(a)** multiple reflections	Mirrors.
10	**(d)** dentist	Magnified image.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Left and right sides appear interchanged (2).	2
12	Regular (1); Diffuse (1).	1+1
13	Distance from pole to principal focus (2).	2
14	A toy using mirrors that forms patterns (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	r = 30 degrees (1); Angle between = 60 degrees (1); Both measured from the normal (1).	1+1+1
16	Spectacles (1); Camera (1); Microscope (1).	1+1+1
17	Mirror laterally inverts (1); Reverses the writing again (1); Appears readable (1).	1+1+1
	OR: i = 0 (1); r = 0 (1); Ray returns along its path (1).	1+1+1
18	Shape (1); Action on rays (1); Uses (1).	1+1+1
	OR: Pole (1); Centre of curvature (1); Principal focus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube with two mirrors (1); 45 degrees (1); Light reflects twice (1); Used in submarines (1); Allows seeing over obstacles (1).	1+1+1+1+1
	OR: Hold lens in sunlight (1); Screen (1); Sharp spot (1); Measure (1); Small bright spot (1).	1+1+1+1+1
20	Convex: converge (1); Concave: diverge (1); Convex: magnifier (1); Concave: spectacles (1); Shapes (1).	1+1+1+1+1
	OR: Myopia: concave (1); Hypermetropia: convex (1); Image on retina (1); Power of lens (1); Prescription (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.